



# SAM 2026

Statistical Analyses  
of Multi-Outcome data  
Bordeaux, France

Monday  
June 29<sup>th</sup>

Morning



BORDEAUX  
POPULATION  
HEALTH

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## One invited KEYNOTE Speech

Amphi Louis

9:15 - 10:15

**Navigating the Crossroads of Statistics, Generative AI, and Genomic Health**

Xihong Lin, Biostatistics, Harvard T.H. Chan School of Public Health

## Session 1

Amphi 10

**Deep learning in survival analysis**

Organizer: Lei Liu - lei.liu@wustl.edu

10:45 - 11:10

**Deep Learning with Time-to-event Outcomes**

Jon Steingrimsson - jon\_steingrimsson@brown.edu

11:10 - 11:35

**Neural Network Machine Regression (NNMR): A Deep Learning Framework for Uncovering High-order Synergistic Effects**

Peter X Song - pxsong@umich.edu

11:35 - 12:00

**Flexible Deep Learning Techniques for Survival Analysis with Data Integration**

Kevin He - kevinhe@umich.edu

## Session 2

Amphi Louis

**Early or intercurrent events in clinical trials**

Organizer: Catherine Legrand

catherine.legrand@uclouvain.be

10:45 - 11:10

**Joint Modeling of Longitudinal Health-Related Quality of Life Data in the Presence of Competing Dropout Risks**

Philippe Lambert - p.lambert@uliege.be

11:10 - 11:35

**A Joint Multi-State Model for Causal Mediation Analysis adjusting for treatment switching in Oncology**

Georgios Kazantzidis - georgios.kazantzidis@roche.com

11:35 - 12:00

**Estimation of inverse probability of censoring weights using shared parameter joint models for longitudinal and survival outcomes.**

Luc Boone - luc.boone@eortc.org

12:00 - 12:25

**Early outcome-based interim analyses in randomized clinical trials with long-term clinical endpoints**

Tomasz Burzykowski - tomasz.burzykowski@iddi.com

## Session 3

Amphi 11

**Trustworthy AI and Statistical Innovation for Complex Health Data**

Organizer: Ying Wei - yw2148@cumc.columbia.edu and TBD

10:45 - 11:10

**Time-varying Quantile Regression with Multi-outcome Latent Groups**

Ying Wei - yw2148@cumc.columbia.edu

11:10 - 11:35

**Quantifying Fidelity in AI Persona Simulations**

Kaizheng Wang - kaizheng.wang@columbia.ca

11:35 - 12:00

**Trajectory Clustering via Spatial-Graph and LLM-Semantic Embeddings**

Nan Lin - nlin@wustl.edu

12:00 - 12:25

~~A Semiparametric Method for Addressing Under-Diagnosis Using Electronic Health Record Data~~

~~Jinbao Chen - jinboche@upenn.edu~~

## Session 4

Amphi 12

**Joint models with Bayesian inference**

Organizer: Sangita Kulathinal

sangita.kulathinal@helsinki.fi

10:45 - 11:10

**Bayesian estimating Transition Rates in Two-State Non-Homogeneous Markov Jump Processes with Intermittent Observations: An Honest Time Data Augmentation Approach**

Etienne Sebag - etienne.sebag@helsinki.fi

11:10 - 11:35

**A Bayesian partition model for multivariate functional data: discovering biological pathways by modeling the time course of gene expression level**

Marion Kerioui - marion.kerioui@inserm.fr

11:35 - 12:00

**Double inverse probability weighting for modeling sparsely measured longitudinal markers in presence of left and right truncation by death**

Trinh Dong Huu - trinh.dong@ncl.ac.uk

12:00 - 12:25

**Robust dynamic borrowing designs for randomized basket trials: a case study from the ultra-rare invasive mold infections**

Satrajit Roychoudhury - satrajit.roychoudhury@pfizer.com



**Session 5**

**Amphi 11**

**Some Advanced Statistical Learning Methods for Modern Scientific Data**

Organizer: Lexin Li - [lexinli@berkeley.edu](mailto:lexinli@berkeley.edu) and chair TBD

Chair: Peter Song - [pxsong@umich.edu](mailto:pxsong@umich.edu)

**13:30 - 13:55**

**Modeling non-uniform hypergraphs using determinantal point processes**

Ji Zhu - [jizhu@umich.edu](mailto:jizhu@umich.edu)

**13:55 - 14:20**

**Structure-aware model-free algorithms for continuous-time reinforcement learning**

Yuhua Zhu - [yuhua.zhu@stat.ucla.edu](mailto:yuhua.zhu@stat.ucla.edu)

**14:20 - 14:45**

**Covariate-adjusted topological inference and learning**

Moo Kyung Chung - [mkchung@wisc.edu](mailto:mkchung@wisc.edu)

**14:45 - 15:10**

**Affiliation: Ranking inference for human feedback tuning in large language models**

Junwei Lu - [junweilu@hsph.harvard.edu](mailto:junweilu@hsph.harvard.edu)

**Session 6**

**Amphi Louis**

**Joint models for risk prediction according to marker variability**

Organizer: Helene Jacqmin-Gadda

[helene.jacqmin-gadda@u-bordeaux.fr](mailto:helene.jacqmin-gadda@u-bordeaux.fr)

**13:30 - 13:55**

**Flexible Bayesian semi-parametric approaches for modelling within-individual variability in joint models**

Jessica Barrett - [jessica.barrett@mrc-bsu.cam.ac.uk](mailto:jessica.barrett@mrc-bsu.cam.ac.uk)

**13:55 - 14:20**

**Joint models with heteroscedastic residual variance: an application to the impact of blood pressure variability on competing health events**

Leonie Courcoul - [leonie.courcoul@u-bordeaux.fr](mailto:leonie.courcoul@u-bordeaux.fr)

**14:20 - 14:45**

**Measuring Biomarker Variability for Survival Data**

Jianxin Pan, China - [jianxinpan@uic.edu.cn](mailto:jianxinpan@uic.edu.cn)

**14:45 - 15:10**

**Joint Modeling of Multiple Longitudinal Biomarkers and Survival Outcomes via Threshold Regression: Variability as a Predictor**

Michael Elliott - [mrelliot@umich.edu](mailto:mrelliot@umich.edu)

**Session 7**

**Amphi 10**

**Recent Developments in Survival Analysis**

Organizer: Yifan Cui - [yifanc095@gmail.com](mailto:yifanc095@gmail.com)

**13:30 - 13:55**

**On an extension of the Cox model for time-dependent covariates under dependent censoring with unknown association**

Ingrid Van Keilegom - [ingrid.vankeilegom@kuleuven.be](mailto:ingrid.vankeilegom@kuleuven.be)

**13:55 - 14:20**

**Semiparametric Regression Analysis of Interval-Censored Data**

Danyu Lin - [lin@bios.unc.edu](mailto:lin@bios.unc.edu)

**14:20 - 14:45**

**Simultaneous Confidence Bands for Time-Varying Hazard Ratios with Interval-Censored Data**

Donglin Zeng - [dzeng@umich.edu](mailto:dzeng@umich.edu)

**14:45 - 15:10**

**Adaptive Mixture-of-Experts Transformers for Joint Modeling of Irregular Longitudinal and Survival Data**

Zhe Fei - [zhfe@ucr.edu](mailto:zhfe@ucr.edu)

**Session 8**

**Amphi 12**

**Trustworthy analysis of healthcare and medical data**

Organizer: Fang Liu - [Fang.Liu.131@nd.edu](mailto:Fang.Liu.131@nd.edu)

and Boris Hejblum

[boris.hejblum@u-bordeaux.fr](mailto:boris.hejblum@u-bordeaux.fr)

**13:30 - 13:55**

**An Agentic AI System for Multi-Framework Communication Coding**

Chuan Hong - [chuan.hong@duke.edu](mailto:chuan.hong@duke.edu)

**13:55 - 14:20**

**Mathematical Foundations of Generative AI for Human Genetics and Epigenetics**

Xinghua Mindy Shi - [mindyshi@temple.edu](mailto:mindyshi@temple.edu)

**14:20 - 14:45**

**Towards Differentially Private Finite Population Estimation: An Approach Based on Survey Weight Regularization**

Yajuan Si - [yajuan@umich.edu](mailto:yajuan@umich.edu)

**14:45 - 15:10**

**A Differentially Private Weighted Empirical Risk Minimization Procedure and its Application to Outcome Weighted Learning**

Fang Liu - [Fang.Liu.131@nd.edu](mailto:Fang.Liu.131@nd.edu)

**Session 9**

**Amphi 11**

**Recent developments in reinforcement learning methods for precision medicine**

Organizer: Donglin Zeng - [dzeng@umich.edu](mailto:dzeng@umich.edu)

**15:40 - 16:05**

**DoubleGen: Debiased Generative Modeling of Counterfactuals**

Alex Luedtke - [aluedtke@uw.edu](mailto:aluedtke@uw.edu)

16:05- 16:30

**Model-free dynamic treatment regimes with arbitrary number of treatments and stages**

*Nilanjana Laha - nilanjanaaa.laha@gmail.com*

16:30- 16:55

**Learning Optimal Early Decision Treatment Rules with Multi-domain Intermediate Outcomes**

*Yuanjia Wang - yw2016@cumc.columbia.edu*

## Session 10

Amphi Louis

**Distributional and Quantile Regressions for complex response data**

Organizer: Antoine Barbieri

*antoine.barbieri@u-bordeaux.fr*

15:40- 16:05

**On asymmetric Laplace regression models: application to trophic diversity**

*Angelo Alcaraz - angelo.alcaraz@ens-paris-saclay.fr*

16:05- 16:30

**Efficient Estimation in Quantile Mixed Models via Smooth Check-Loss Approximation**

*Lei Liu - lei.liu@wustl.edu*

16:30- 16:55

**A joint model based on distributional regression to study the link between blood pressure and the risk of cerebral vasospasm.**

*Mouna Abed - mouna.abed@u-bordeaux.fr*

16:55- 17:20

**Optimal-transport based conformal prediction**

*Gauthier Thurin - Gauthier.thurin@ens.fr*

## Session 11

Amphi 12

**Innovative Statistical Methods for Complex Health and Biomedical Data**

Organizer: Mei Hao - hao.mei@ruc.edu.cn

15:40- 16:05

~~**A Performance-Based Framework for Transfer Learning Measurement and Guidance**~~

~~*Hao Zhang - haozhang@arizona.edu*~~

16:05- 16:30

**BELIEF in Dependence**

*Kai Zhang - zhangk@email.unc.edu*

16:30- 16:55

**Brain Encoding and Decoding: Some Examples**

*Lexin Li - lexinli@berkeley.edu*

16:55- 17:20

**Functional Latent Space Model for Time-Varying Networks: Application to Taiwan Administrative Claims Data**

*Hao Mei - hao.mei@ruc.edu.cn*

## Session 12

Amphi 10

**High-dimensional surrogate markers in multi-outcome settings**

Organizer: Boris Hejblum

*boris.hejblum@u-bordeaux.fr*

15:40- 16:05

**Resilience Measures for the Surrogate Paradox**

*Layla Parast - parast@austin.utexas.edu*

16:05- 16:30

**Meta-Analytic Evaluation of Complex Surrogate Endpoints based on the Surrogate Index**

*Florian Stijven - florian.stijven@kuleuven.be*

16:30- 16:55

**Stable Multi-Surrogate Transformation for Robust and Generalizable Surrogacy**

*Tianxi Cai - tcai@hsph.harvard.edu*

16:55- 17:20

**Surrogate endpoint evaluation for outcomes with high-dimensional features, with application to vaccines**

*Peter Gilbert - peterg@uw.edu*

Monday June 29<sup>th</sup>

Afternoon







**Session 13**

**Amphi 12**

**Recent advances in transfer learning for biomedical applications**

Organizer: Lan Luo - [ll1118@sph.rutgers.edu](mailto:ll1118@sph.rutgers.edu)

**9:00 - 09:25**

**Unsupervised Aggregation of Multiple Learning Algorithms**

Rui Duan - [rduan@hsph.harvard.edu](mailto:rduan@hsph.harvard.edu)

**9:25 - 09:50**

**Heterogeneity-adaptive meta-analysis**

Emily C Hector - [ehector@umich.edu](mailto:ehector@umich.edu)

**9:50 - 10:15**

**Hierarchical Projection for Adaptive Knowledge Transfer**

Tian Gu - [tg2880@cumc.columbia.edu](mailto:tg2880@cumc.columbia.edu)

**10:15 - 10:40**

~~**On a synergistic learning phenomenon in nonparametric domain adaptation**~~

~~Ling Zhou - [zhouling@swufe.edu.cn](mailto:zhouling@swufe.edu.cn)~~

**Session 14**

**Amphi Louis**

**Clustering from multivariate data**

Organizer: Anais Rouanet

[anais.rouanet@u-bordeaux.fr](mailto:anais.rouanet@u-bordeaux.fr) and

Julie Fendler

[julie.fendler@mrs-bsu.cam.ac.uk](mailto:julie.fendler@mrs-bsu.cam.ac.uk)

**9:00 - 09:25**

**Clustering of Multivariate Longitudinal Data of mixed type**

Julien Jacques - [julien.jacques@univ-lyon2.fr](mailto:julien.jacques@univ-lyon2.fr)

**9:25 - 09:50**

**Consensus Monte Carlo for mixtures of categorical distributions**

Julie Fendler - [julie.fendler@mrc-bsu.cam.ac.uk](mailto:julie.fendler@mrc-bsu.cam.ac.uk)

**9:50 - 10:15**

**Non-parametric clustering of multivariate longitudinal data with variable selection: Identifying sub-phenotypes of Alzheimer's Disease**

Anais Rouanet - [anais.rouanet@u-bordeaux.fr](mailto:anais.rouanet@u-bordeaux.fr)

**10:15 - 10:40**

**A Dynamic Joint Latent Class Model With Time-Dependent Transition Probabilities**

Gendresa Selimi - [gendresa.selimi@manchester.ac.uk](mailto:gendresa.selimi@manchester.ac.uk)

**Session 15**

**Amphi 10**

**The use of functional data analysis in multivariate settings**

Organizer: Helene Jacqmin-Gadda

[helene.jacqmin-gadda@u-bordeaux.fr](mailto:helene.jacqmin-gadda@u-bordeaux.fr)

**9:00 - 09:25**

**Functional joint model for longitudinal and survival data with a functional covariate with application to a longitudinal study of objective physical activity**

Luo Xiao - [lxiao5@ncsu.edu](mailto:lxiao5@ncsu.edu)

**9:25 - 09:50**

**Estimation and Application of Derivative Multivariate Functional Principal Component Analysis**

Andrew Simpkin - [andrew.simpkin@universityofgalway.ie](mailto:andrew.simpkin@universityofgalway.ie)

[universityofgalway.ie](mailto:universityofgalway.ie)

**9:50 - 10:15**

**Random Survival Forest for the prediction of survival outcomes from time-varying outcomes using functional principal component analysis**

Robin Genuer - [robin.genuer@u-bordeaux.fr](mailto:robin.genuer@u-bordeaux.fr)

**10:15 - 10:40**

~~**Preliminary Test Estimation in ULAN models for Functional Data**~~

~~Sophie Dabo - [sophie.dabo@univ-lille.fr](mailto:sophie.dabo@univ-lille.fr)~~

**Session 16**

**Amphi 11**

**New advances in survival analysis and machine learning**

Organizer: Yichuan Zhao - [yichuan@gsu.edu](mailto:yichuan@gsu.edu) and

Liang Li - [lli15@mdanderson.org](mailto:lli15@mdanderson.org)

**9:00 - 09:25**

**Prediction Machines Based on Concordance Correlation**

Edsel Pena - [pena@stat.sc.edu](mailto:pena@stat.sc.edu)

**9:25 - 09:50**

**Double Machine Learning of Continuous Treatment Effects with General Instrumental Variables**

Yifan Cui - [yifanc095@gmail.com](mailto:yifanc095@gmail.com)

**9:50 - 10:15**

**Modeling the Association Between Multivariate Nonlinear Longitudinal Data and Survival Outcome: A Measurement Error Perspective**

Liang Li - [lli15@mdanderson.org](mailto:lli15@mdanderson.org)

**10:15 - 10:40**

**Novel empirical likelihood method for the cumulative hazard ratio under stratified Cox models**

Yichuan Zhao - [yichuan@gsu.edu](mailto:yichuan@gsu.edu)

**Session 17**

**Amphi 11**

**Statistical Learning for Complex Data**

Organizer: Xingqiu Zhao - [xingqiu.zhao@polyu.edu.hk](mailto:xingqiu.zhao@polyu.edu.hk)

Chair: Yu Gu - [yugu@hku.hk](mailto:yugu@hku.hk)

**11:10 - 11:35**

**Semiparametric Inference for Functional Survival**

Qixian Zhong - [qxzhong@xmu.edu.cn](mailto:qxzhong@xmu.edu.cn)

**11:35 - 12:00**

**Deep Nonparametric Inference for Interval-Censored Data**

Qiang Wu - [qiangwu@mail.bnu.edu.cn](mailto:qiangwu@mail.bnu.edu.cn)

**12:00 - 12:25**

**Deep Conditional Generative Learning for Optimal Individualized Treatment Rules**

Xiangbin Hu - [xiang-bin.hu@connect.polyu.hk](mailto:xiang-bin.hu@connect.polyu.hk)

**Session 18**

**Amphi Louis**

**Software for the Analysis of Multi-Outcome Models (INLAjoint, JMbayer2, frailtypack ...)**

Organizer: Denis Rustand

[denis.rustand@u-bordeaux.fr](mailto:denis.rustand@u-bordeaux.fr)

**11:10 - 11:35**

**Graphpcor: Models for correlation matrices based on graphs**

Elias Krainski - [elias.krainski@kaust.edu.sa](mailto:elias.krainski@kaust.edu.sa)

**11:35 - 12:00**

**Assessing surrogacy from joint modeling and mediation analysis when surrogates are either censored event times or longitudinal biomarker: cancer application : FRAILTYPACK**

Virginie Rondeau - [virginie.rondeau@inserm.fr](mailto:virginie.rondeau@inserm.fr)

**12:00 - 12:25**

**INLAjoint and beyond: INLA for multi-outcome modeling**

Janet van Niekerk - [janet.vanNiekerk@up.ac.za](mailto:janet.vanNiekerk@up.ac.za)

**12:25 - 12:50**

**The JMbayer2 R package for joint models of longitudinal and time-to-event data: From flexible fitting to dynamic prediction and validation**

Pedro Miranda-Afonso

[p.mirandaafonso@erasmusmc.nl](mailto:p.mirandaafonso@erasmusmc.nl)

**Session 19**

**Amphi 10**

**Advanced Analytical Methods in Multi-outcome Data**

Organizer: Lei Liu - [lei.liu@wustl.edu](mailto:lei.liu@wustl.edu)

**11:10 - 11:35**

**Function-on-Scalar Regression with Ultra high-Dimensional Error-Prone Covariates**

Grace Y. Yi - [gyi5@uwo.ca](mailto:gyi5@uwo.ca)

**11:35 - 12:00**

**Time-varying frailty model for analyzing recurrent/terminal event data**

Shalini Ramachandra

[shalini.ramachandra@pennmedicine.upenn.edu](mailto:shalini.ramachandra@pennmedicine.upenn.edu)

**12:00 - 12:25**

**Inference for Deep Neural Network Estimators**

Yi Li - [yili@umich.edu](mailto:yili@umich.edu)

**Session 20**

**Amphi 12**

**Causal Learning for Optimal Health Decision-Making with Complex Outcomes**

Organizer: Yuanjia Wang

[yw2016@cumc.columbia.edu](mailto:yw2016@cumc.columbia.edu)

**11:10 - 11:35**

**Estimating the optimal allocation of kidneys in the presence of competing risks from clustered data**

Erica Moodie - [erica.moodie@mcgill.ca](mailto:erica.moodie@mcgill.ca)

**11:35 - 12:00**

**A structural nested rate model for estimating the effects of time-varying exposure on recurrent event outcomes**

Ashken Ertefie - [Ashkan.Ertefaie@PennMedicine.upenn.edu](mailto:Ashkan.Ertefaie@PennMedicine.upenn.edu)

**12:00 - 12:25**

**Learning Locally Interpretable Individualized Treatment Rules**

Yuan Chen - [cheny19@mskcc.org](mailto:cheny19@mskcc.org)

**12:25 - 12:50**

**Joint Model for Mediation Analysis with Causally Related Longitudinal and Recurrent Event Mediators for Survival Outcome**

Cheng Zheng - [cheng.zheng@unmc.edu](mailto:cheng.zheng@unmc.edu)

**Tuesday June 30<sup>th</sup>**

**Morning**





# SAM 2026

Statistical Analyses  
of Multi-Outcome data  
Bordeaux, France

**Tuesday**  
**June 30<sup>th</sup>**

Afternoon



BORDEAUX  
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## Session 21

Amphi 10

### Statistical and machine learning methods for complex biomedical data

Organizer: QI Long - qlong@pennmedicine.upenn.edu

Chair: Kevin He - kevinhe@umich.edu

14:00 - 14:25

#### A Robust Local Frechet Regression Using Unbalanced Neural Optimal Transport with Applications to Dynamic Single-cell Genomics Data

Hongzhe Li - hongzhe@upenn.edu

14:25 - 14:50

#### iDeepViewLearn+: Interpretable AI for Robust and Fair Multimodal Data Integration

Sandra Safo - ssafo@umn.edu

14:50 - 15:15

#### Ranking-based Data Integration for Robust Outcomes Prediction and Biomarker Detection

Nicholas Henderson - nchender@umich.edu

## Session 22

Amphi Louis

### Causal inference in presence of truncation by death

Organizer: Cecile Proust-Lima

cecile.proust-lima@u-bordeaux.fr

14:00 - 14:25

#### Causal effects on non-terminal event time with application to antibiotic usage and future resistance

Daniel Nevo - danielnevo@gmail.com

14:25 - 14:50

#### Bias-Corrected Two-Stage Modeling for Joint Analysis of Multidimensional HRQoL and Survival

Catherine Legrand - catherine.legrand@uclouvain.be

14:50 - 15:15

#### A path-specific effect approach to mediation analysis with time-varying mediators and time-to-event outcomes accounting for competing risks

Arce Domingo-Relloso - ad3531@cumc.columbia.edu

15:15 - 15:40

#### Evaluating Treatment Effects in Longitudinal Trials with Truncation by Death: An Assumption-Lean Approach

Stijn Vansteelandt - stijn.vansteelandt@ugent.be

## Session 23

Amphi 11

### Innovative Statistical Methods for Longitudinal and Survival Analysis in Medical Studies

Organizer: Yanqing Sun - yasun@charlotte.edu

14:00 - 14:25

#### Joint models in longitudinal studies for efficient and robust inferences

Lang Wu - lang@stat.ubc.ca

14:25 - 14:50

#### Enhanced survival trees with applications to Alzheimer's disease study

Jimin Ding - jmding@wustl.edu

14:50 - 15:15

#### Boosting methods for interval censored data with regression and classification

Wenqing He - whe@stats.uwo.ca

15:15 - 15:40

#### Estimation and Inference of Semiparametric Temporal Intensity Models for Recurrent Events with Application to a Malaria Vaccine Trial

Yanqing Sun - yasun@charlotte.edu

## Session 24

Amphi 12

### New Estimation and Inference Methods for Analyzing Composite Survival Outcomes

Organizer: Ying Ding - yingding@pitt.edu

Chair: Yichuan Zhao - yichuan@gsu.edu

14:00 - 14:25

#### On statistical inference of multiple competing risks in oncology clinical trials

Chen Hu - huc@jhu.edu

14:25 - 14:50

#### Win odds for Sequential Multiple Assignment Randomized Trials

Yu Cheng - yucheng@pitt.edu

14:50 - 15:15

#### Variable Selection for Time-Varying Covariates in Multistate Survival Models with Interval Censoring

Ariane Bercu - ariane.bercu@u-bordeaux.fr

15:15 - 15:40

#### Joint Modeling of True and Misclassified Event Times for Biomarker Evaluation

Yingye Zheng - yzheng@fredhutch.org

## Session 25

Amphi 11

### Machine Learning and Survival Analysis

Organizer: Jiaoguo Sun - sunj@missouri.edu

Chair: Qixian Zhong - qxzhong@xmu.edu.cn

16:10 - 16:35

#### Semiparametric Functional Multi-State Models: Estimation and Testing with Application to Alzheimer's Disease

Yu Gu - yugu@hku.hk

16:35 - 17:00

#### Spline Learnable Neural Networks with Applications to Generative Diffusion Models

Jiaqi Zhao - jiaqi2025.zhao@connect.polyu.hk

17:00 - 17:25

#### Automatic structure identification and variable selection for additive accelerated failure time model



with ultra high dimensional covariates

Li Liu - [liu.math@whu.edu.cn](mailto:liu.math@whu.edu.cn)

17:25 - 17:50

**Deep Semiparametric and Nonparametric Inference for Longitudinal Data**

Rujia Cao - [rujia.cao@connect.polyu.hk](mailto:rujia.cao@connect.polyu.hk)

#### Session 26

Amphi Louis

**Area of causal inference**

Organizer: Cheng Zheng - [cheng.zheng@unmc.edu](mailto:cheng.zheng@unmc.edu)

16:10 - 16:35

**Joint Analysis of Multivariate Longitudinal Data and Interval-Censored Event Time Data with Application to Huntington's Disease Progression**

Ying Zhang - [ying.zhang@unmc.edu](mailto:ying.zhang@unmc.edu)

16:35 - 17:00

**Model-free High Dimensional Mediator Selection with False Discovery Rate Control**

Ran Dai - [ran.dai@unmc.edu](mailto:ran.dai@unmc.edu)

17:00 - 17:25

**Confidence Sets for Causal Orderings**

Mladen Kolar - [mkolar@marshall.usc.edu](mailto:mkolar@marshall.usc.edu)

17:25 - 17:50

**Doubly robust inference for the heterogeneous treatment effect of multiple treatments on time-to-event outcomes**

Youngjoo Cho - [yvc5154@konkuk.ac.kr](mailto:yvc5154@konkuk.ac.kr)

#### Session 27

Amphi 10

**Joint analysis of time-to-event or longitudinal markers in the presence of mortality as a competing risk**

Organizer: Sebastien Haneuse

[shaneuse@hsph.harvard.edu](mailto:shaneuse@hsph.harvard.edu)

16:10 - 16:35

**Challenges and Opportunities in Joint Modelling of Multiple Longitudinal and Survival Outcomes**

Eleni-Rosalina Andrinopoulou

[eleni.r.andrinopoulou@gsk.com](mailto:eleni.r.andrinopoulou@gsk.com)

16:35 - 17:00

**A Bayesian Nonparametric Approach for Semi-Competing Risks with Application to Cardiovascular Health**

Daniels, Michael J - [mdaniels@stat.ufl.edu](mailto:mdaniels@stat.ufl.edu)

17:00 - 17:25

**A novel framework for Joint prediction of longitudinal markers and death**

Sebastien Haneuse - [shaneuse@hsph.harvard.edu](mailto:shaneuse@hsph.harvard.edu)

17:25 - 17:50

**Nonparametric estimation of the joint and conditional survival functions of the time to an event of interest and associated integrated covariate processes**

**Tuesday June 30<sup>th</sup>**

Afternoon

Sangita Kulathinal - [sangita.kulathinal@helsinki.fi](mailto:sangita.kulathinal@helsinki.fi)

#### Session 28

Amphi 12

**Nonparametric and semiparametric modeling for the analysis of complex data**

Organizer: (Zhezhen Jin - [zj7@cumc.columbia.edu](mailto:zj7@cumc.columbia.edu))

Chairs Ariane Bercu - [ariane.bercu@u-bordeaux.fr](mailto:ariane.bercu@u-bordeaux.fr)

16:10 - 16:35

**Robust causal effect estimation in high-dimensional survival analysis using sufficient dimension reduction**

Shanshan Ding - [sding@udel.edu](mailto:sding@udel.edu)

16:35 - 17:00

**Variable Selection in the Joint Frailty Model of Recurrent and Terminal Events Using Broken Adaptive Ridge Regression.**

Xuewen Lu - [xlu@ucalgary.ca](mailto:xlu@ucalgary.ca)

17:00 - 17:25

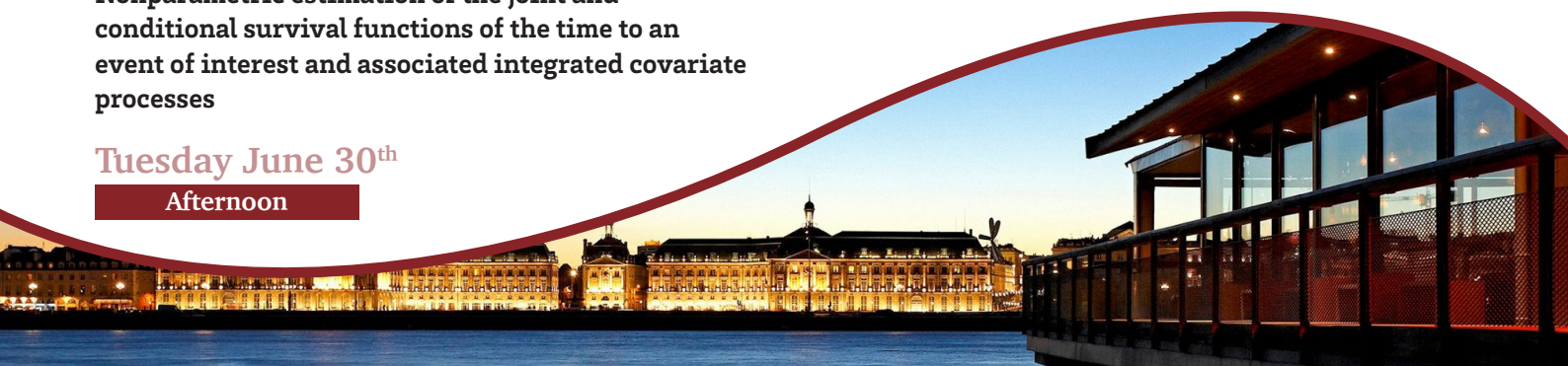
**Modeling Heterogeneous Disease Processes Using a Mixture of Multistate Hidden Markov Models**

Leilei Zeng - [lzeng@uwaterloo.ca](mailto:lzeng@uwaterloo.ca)

17:25 - 17:50

**Integrative Analysis of Multiple Data Sources Involving Misclassification and Missing Data**

Hua Shen - [hua.shen@ucalgary.ca](mailto:hua.shen@ucalgary.ca)





**Session 29**

**Amphi Louis**

**Machine or deep learning for multiple longitudinal and survival data**

Organizer: Virginie Rondeau

[virginie.rondeau@inserm.fr](mailto:virginie.rondeau@inserm.fr)

**9:00 - 09:25**

**Joint models and deep learning neural networks for the longitudinal analysis of mammography images in screen-detected breast cancer risk prediction**

Manel Rakez - [rakez.manel@outlook.fr](mailto:rakez.manel@outlook.fr)

**9:25 - 09:50**

**Deep Learning for Longitudinal and Survival Data: Toward Realistic Synthetic Patient Generation**

Agathe Guilloux - [agathe.guilloux@inria.fr](mailto:agathe.guilloux@inria.fr)

**9:50 - 10:15**

**Modeling disease progression and treatment switches in rare disease trials by integrating statistical modeling and synthetic experts**

Clemens Schaechter - [clemens.schaechter@uniklinik-freiburg.de](mailto:clemens.schaechter@uniklinik-freiburg.de)

**10:15 - 10:40**

**Neural differential equations enhanced linear mixed model**

Cécile Proust-Lima - [cecile.proust-lima@u-bordeaux.fr](mailto:cecile.proust-lima@u-bordeaux.fr)

**Session 30**

**Amphi 11**

**Causal mediation pathway analysis**

Organizer: Peter Song - [pxsong@umich.edu](mailto:pxsong@umich.edu)

**9:00 - 09:25**

**Spillovers Through Time: A Neural Framework for Temporal Network Data and Dynamic Causal Effect Estimation**

Jiasheng Shi - [shijiasheng@cuhk.edu.cn](mailto:shijiasheng@cuhk.edu.cn)

**9:25 - 09:50**

**Semiparametric mediation analysis using single-index models**

Yen-Tsung Huang - [ythuang@stat.sinica.edu.tw](mailto:ythuang@stat.sinica.edu.tw)

**9:50 - 10:15**

**Mitigating Unmeasured Confounding Bias in Large-Scale Causal Mediation Analysis Via Factor Analysis in Epigenome-Wide Studies**

Zhonghua Liu - [zl2509@cumc.columbia.edu](mailto:zl2509@cumc.columbia.edu)

**10:15 - 10:40**

**Statistical Inference for Mediation Models with High Dimensional Exposures and Mediators**

Wei Zhou - [zhouwei23@swufe.edu.cn](mailto:zhouwei23@swufe.edu.cn)

**Session 31**

**Amphi 10**

**Advanced uses of joint modeling techniques with application in longitudinal and time-to-event data**

Organizer: Dimitris Rizopoulos

[d.rizopoulos@erasmusmc.nl](mailto:d.rizopoulos@erasmusmc.nl)

**9:00 - 09:25**

**Dynamic Predictions and Predictimands for Salvage Therapy in Recurrent Prostate Cancer Using Joint Models**

Jeremy Taylor - [jmgt@umich.edu](mailto:jmgt@umich.edu)

**9:25 - 09:50**

**Scalable Time-Dynamic Joint Models for Multilevel Health Data**

Esra Kurum - [esra.kurum@ucr.edu](mailto:esra.kurum@ucr.edu)

**9:50 - 10:15**

**Shared parameter modeling of longitudinal data allowing for possibly informative visiting process and terminal event**

Christos Thomadakis - [cthomadak@aueb.gr](mailto:cthomadak@aueb.gr)

**10:15 - 10:40**

**Addressing measurement times that are not at random in longitudinal data through joint modelling of outcomes and latent disease processes**

Eleanor Pullenayegum

[eleanor.pullenayegum@sickkids.ca](mailto:eleanor.pullenayegum@sickkids.ca)

**Session 32**

**Amphi 12**

**Tensor-Train Methods, Partial Derivatives, and High-dimensional Inference in Modern Statistics**

Organizer: Yong Chen

[ychen123@pennmedicine.upenn.edu](mailto:ychen123@pennmedicine.upenn.edu)

**9:00 - 09:25**

**PDA: a unified framework for lossless, one-shot, federated learning algorithms**

Yong Chen - [ychen123@pennmedicine.upenn.edu](mailto:ychen123@pennmedicine.upenn.edu)

**9:25 - 09:50**

**Projection-Enhanced Interpolation-Based Tensor-Train Decompositions for One-Shot, Privacy-Preserving Distributed Inference**

Jingmei Qiu - [jingqiu@udel.edu](mailto:jingqiu@udel.edu)

**9:50 - 10:15**

**Statistical Learning via Partial Derivatives**

Xiaowu Dai - [dai@stat.ucla.edu](mailto:dai@stat.ucla.edu)

**10:15 - 10:40**

**High-dimensional Data Assimilation for Large Scale Physical Systems**

Yumou Qiu - [qiuyumou@math.pku.edu.cn](mailto:qiuyumou@math.pku.edu.cn)

**One invited KEYNOTE Speech**

**Amphi Louis**

**11:10 - 12:10**

**Harnessing Variational Auto-Encoders for Fitting Deep Mixed-Effects Models**

Dimitris Rizopoulos, Erasmus University Medical Center - [d.rizopoulos@erasmusmc.nl](mailto:d.rizopoulos@erasmusmc.nl)